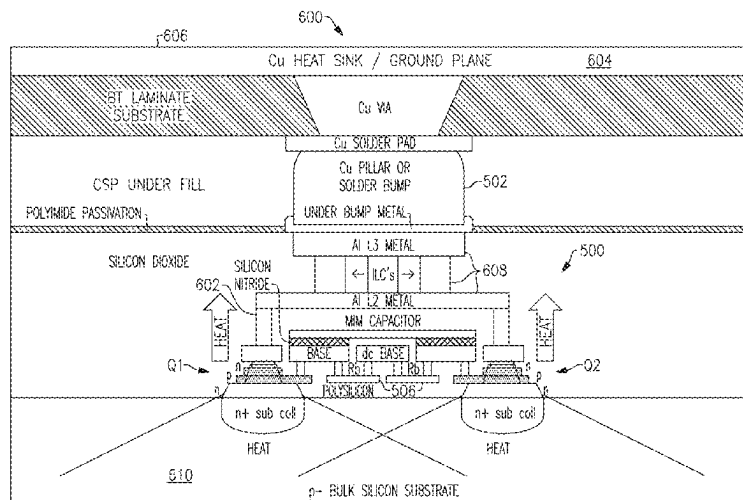


(45) **Date of Patent:** *Sep. 16, 2025

- 22 Claims, 15 Drawing Sheets**



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continuation of application No. 16/056,251, filed on Aug. 6, 2018, now Pat. No. 10,790,788, which is a continuation of application No. 15/299,738, filed on Oct. 21, 2016, now Pat. No. 10,069,466.

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H04B 1/48 (2006.01)
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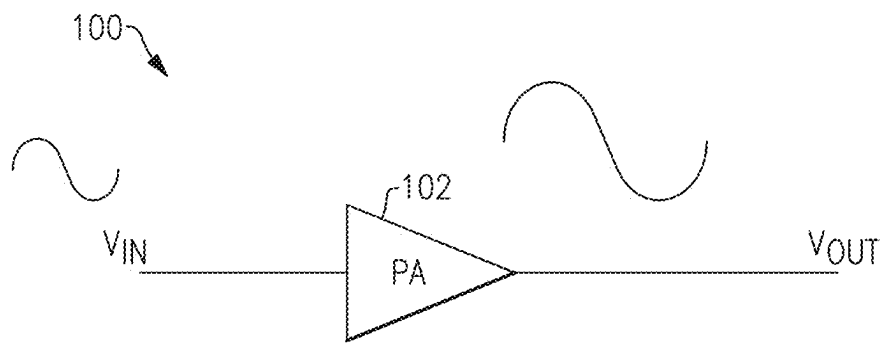
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FIG.1

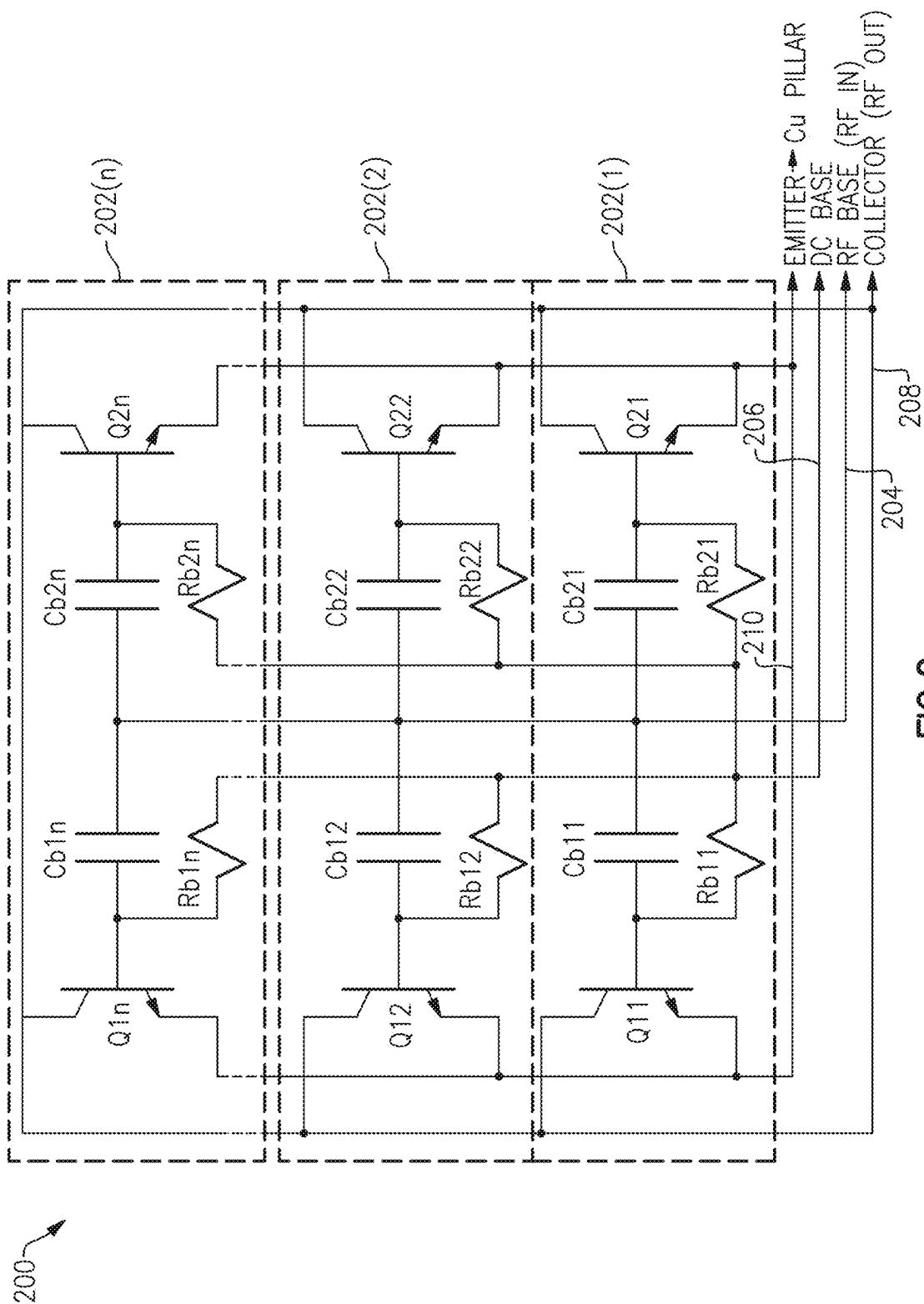


FIG. 2

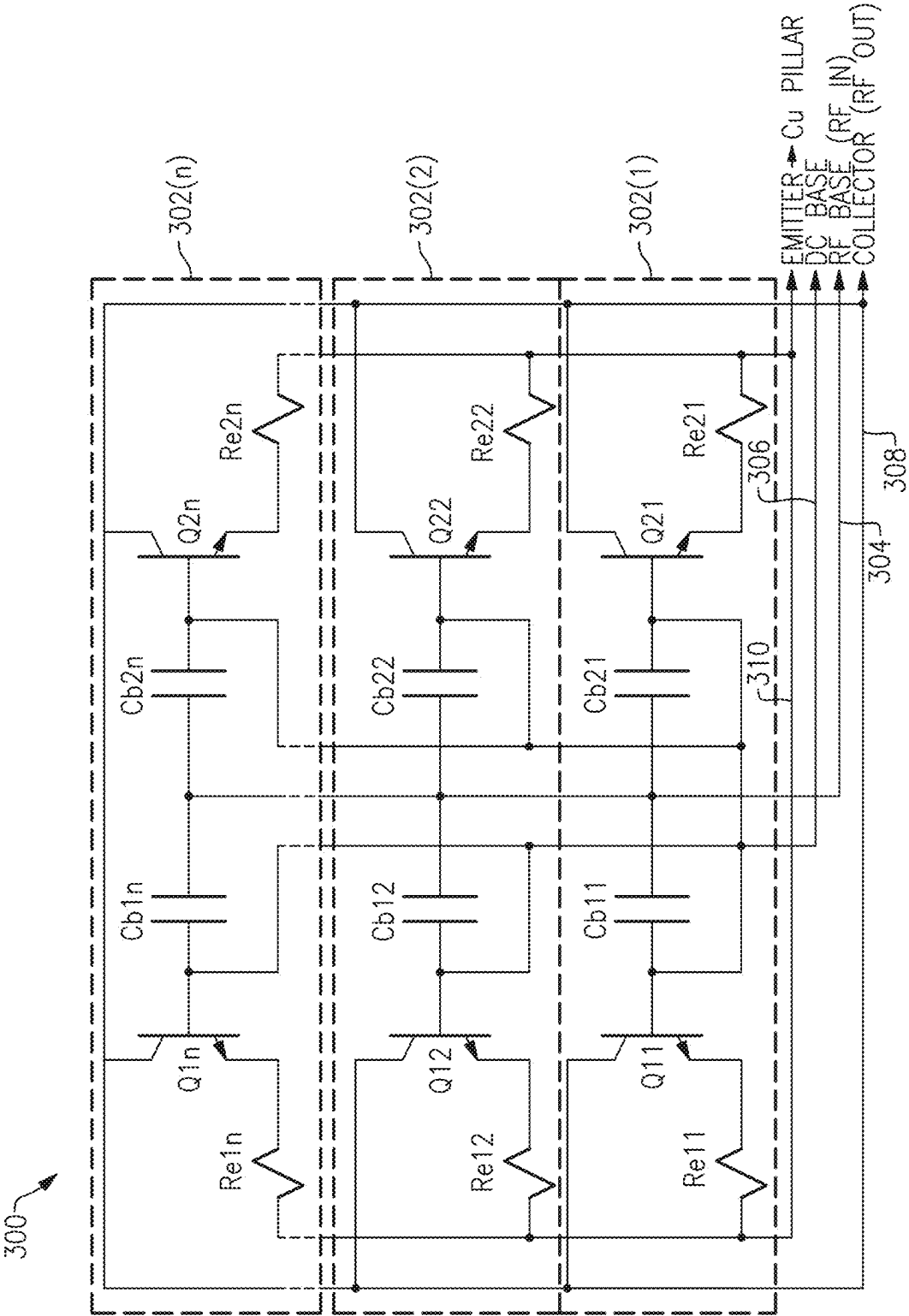
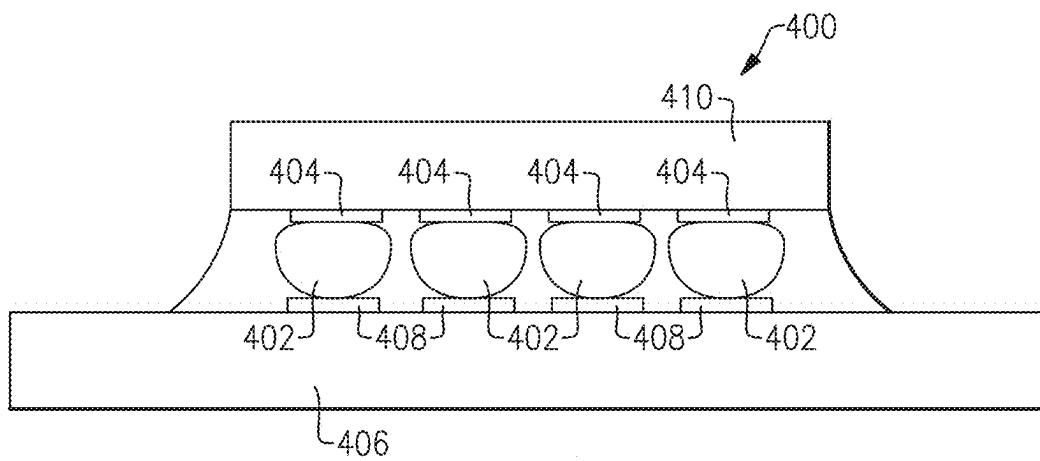


FIG.3

**FIG. 4**

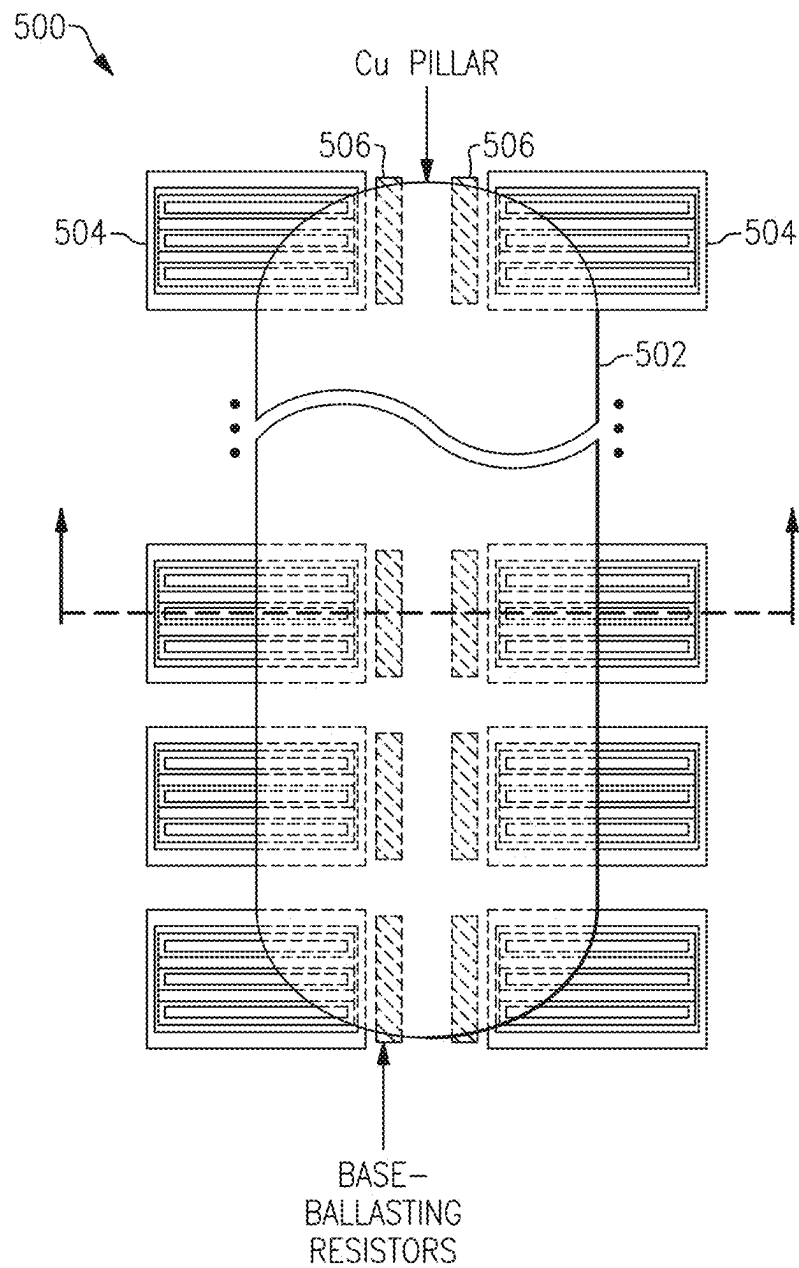


FIG. 5

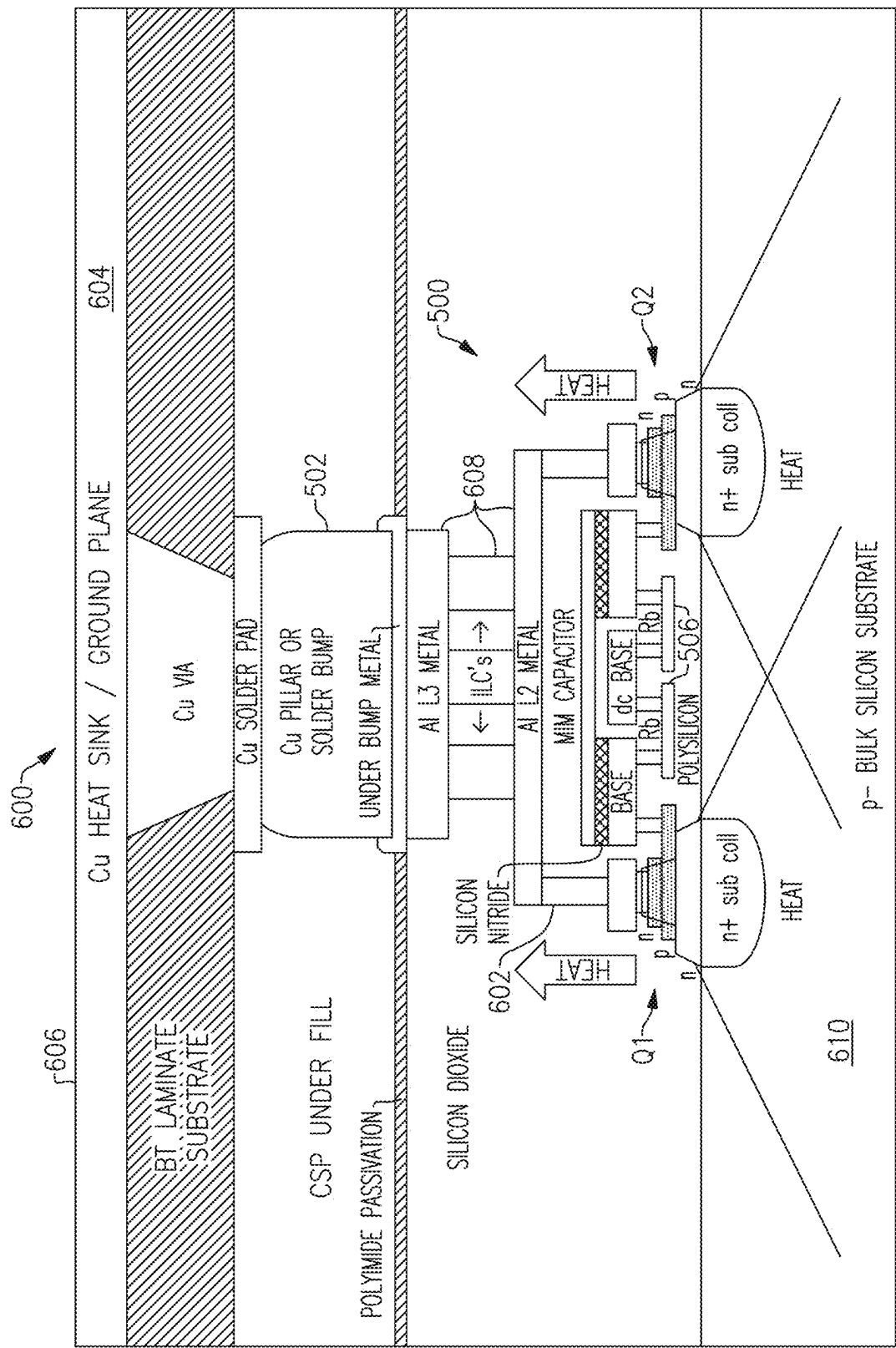
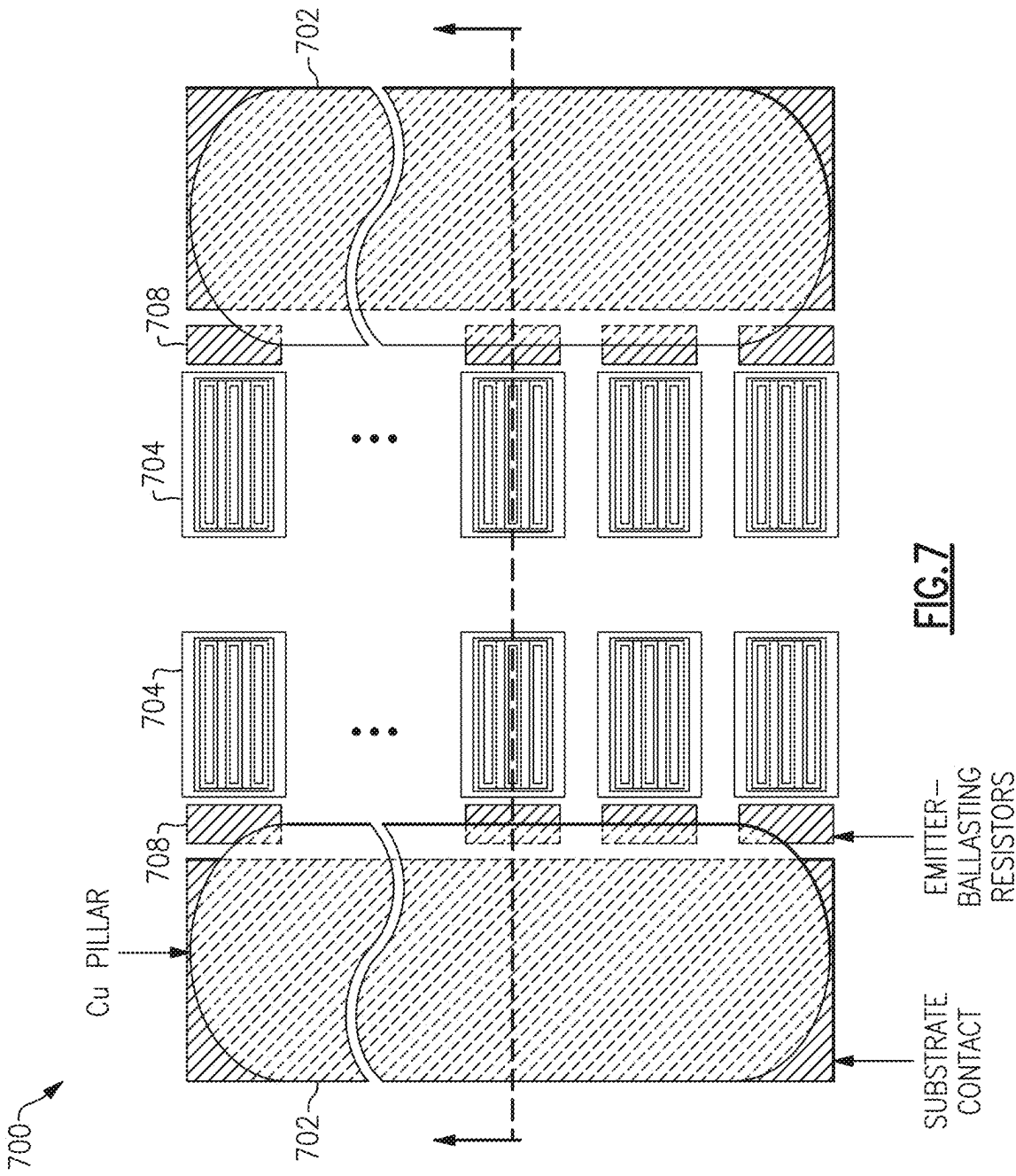


FIG.6



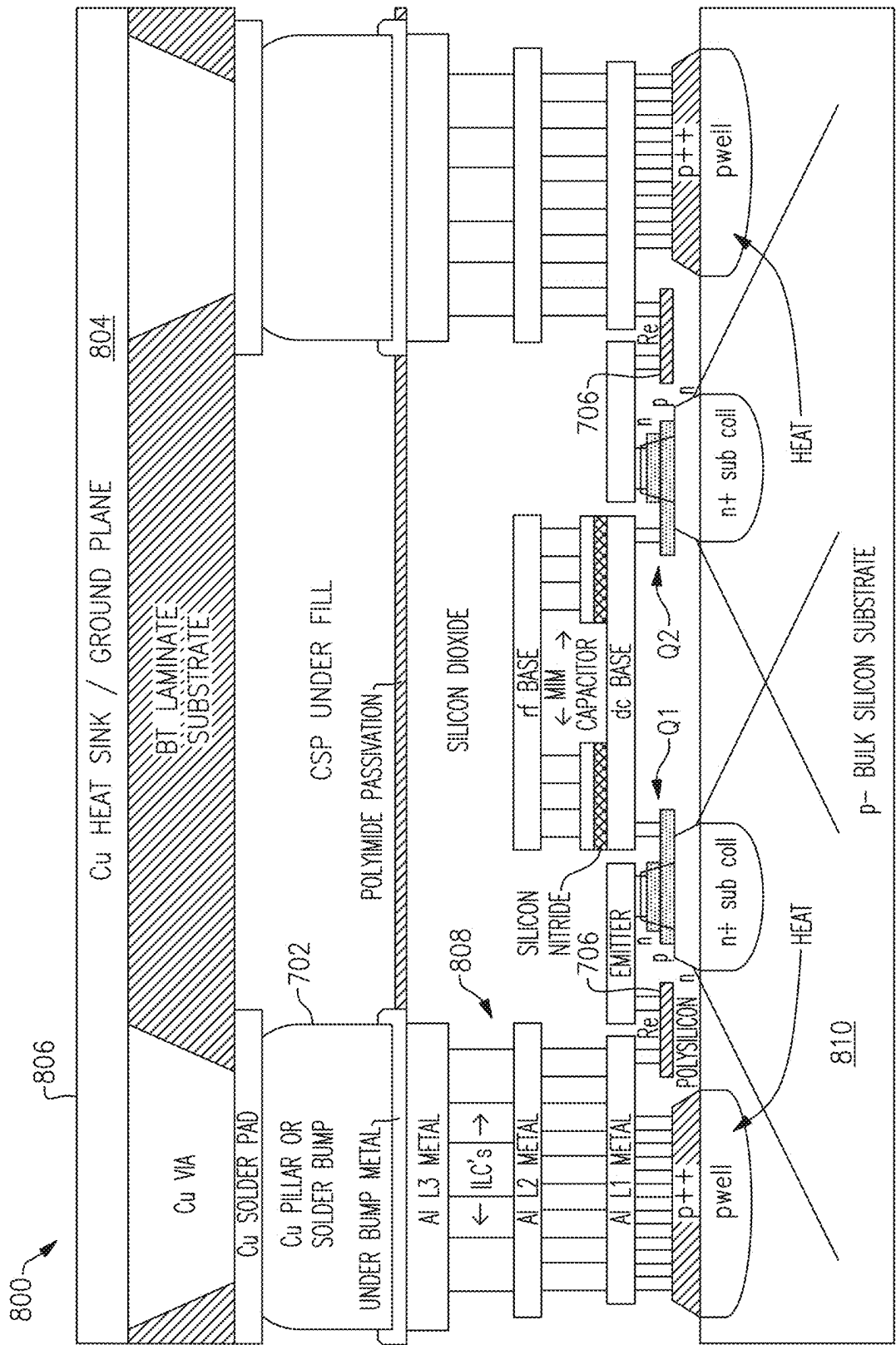
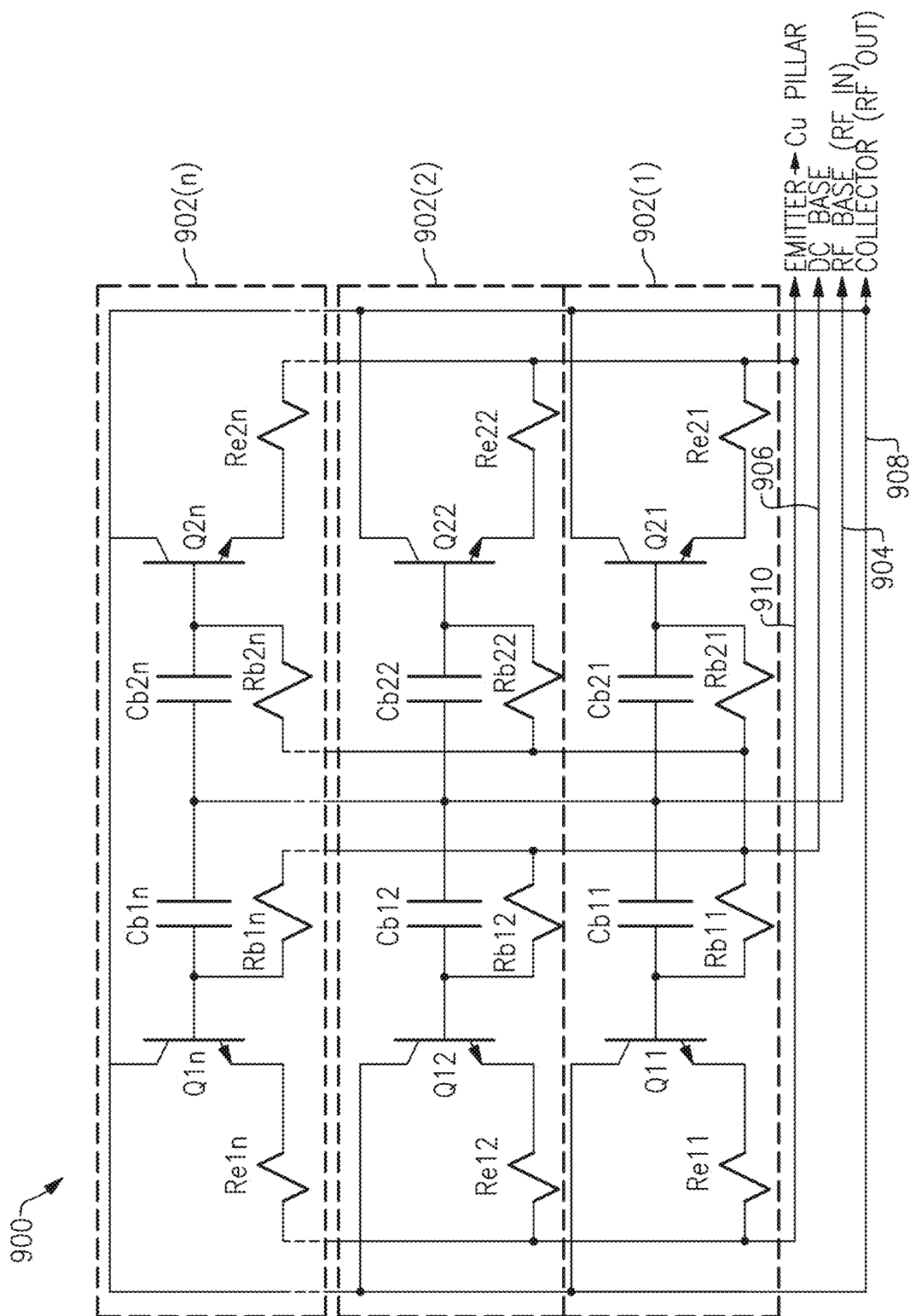
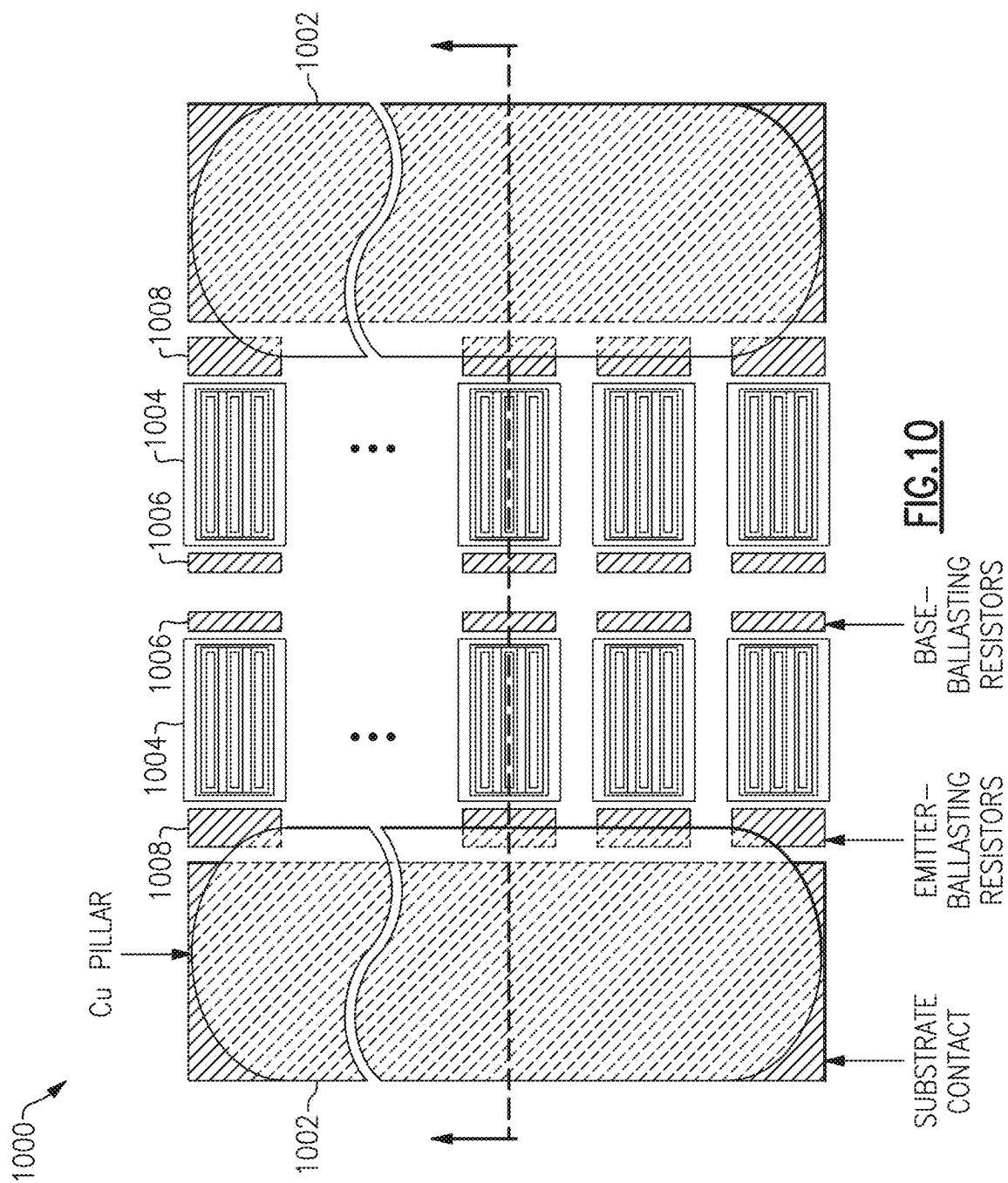


FIG. 8





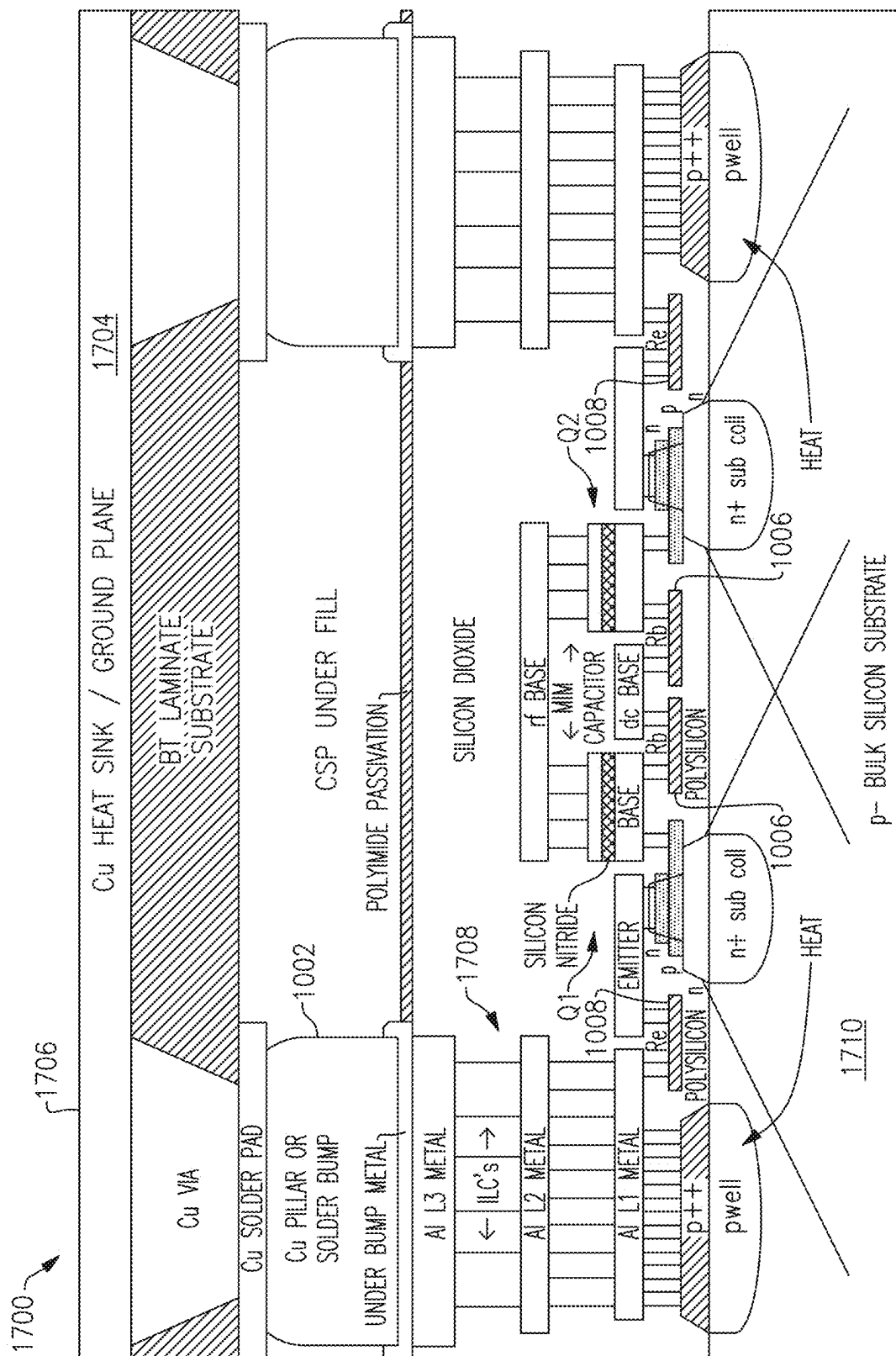


Fig.

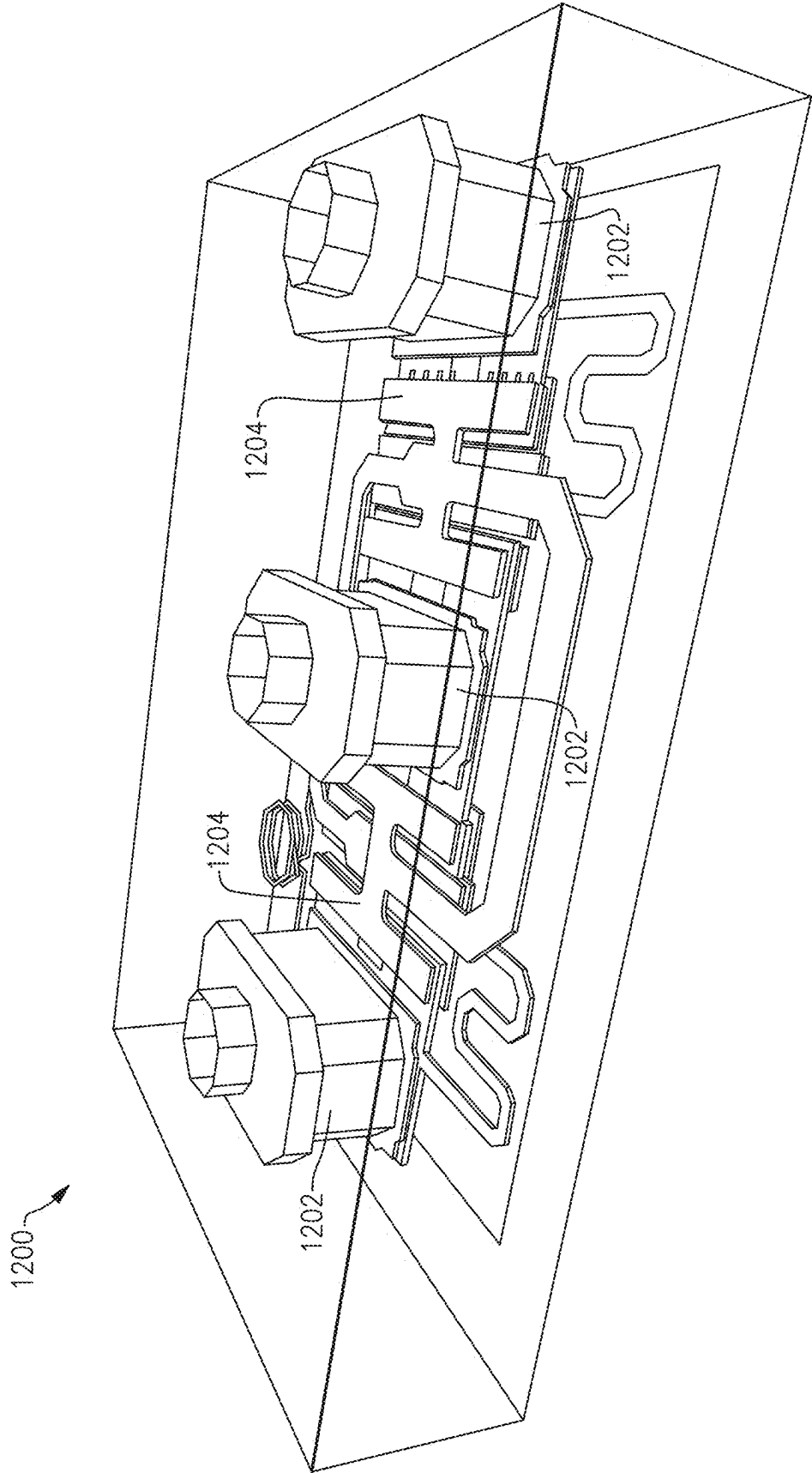
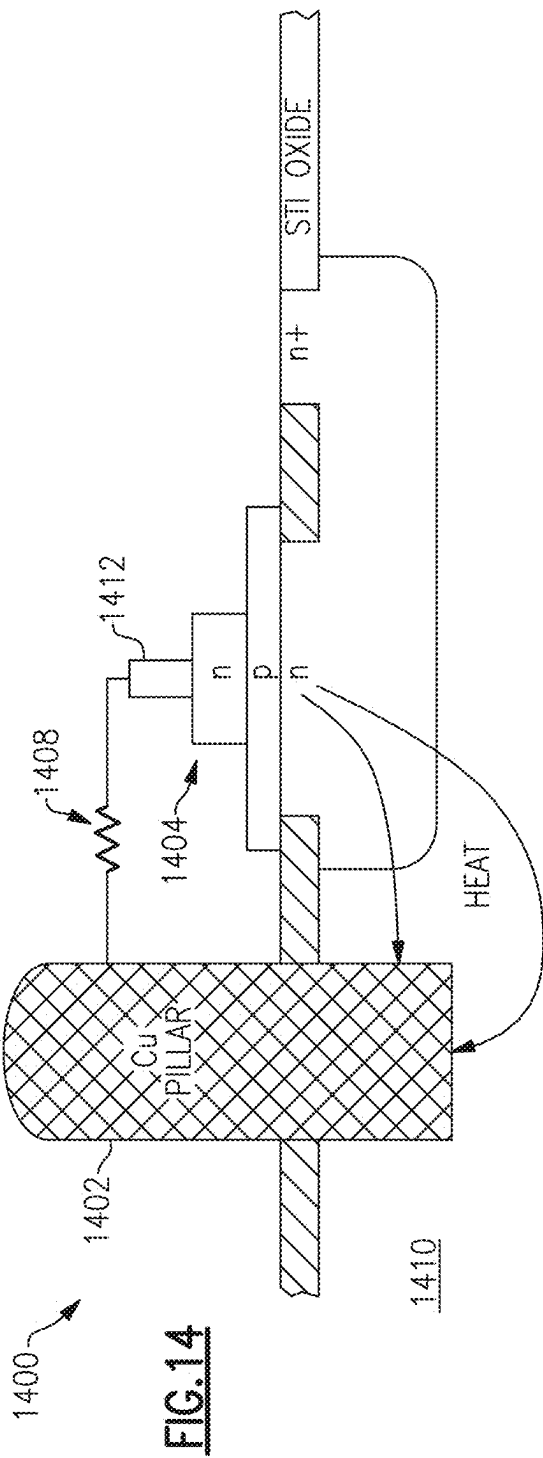
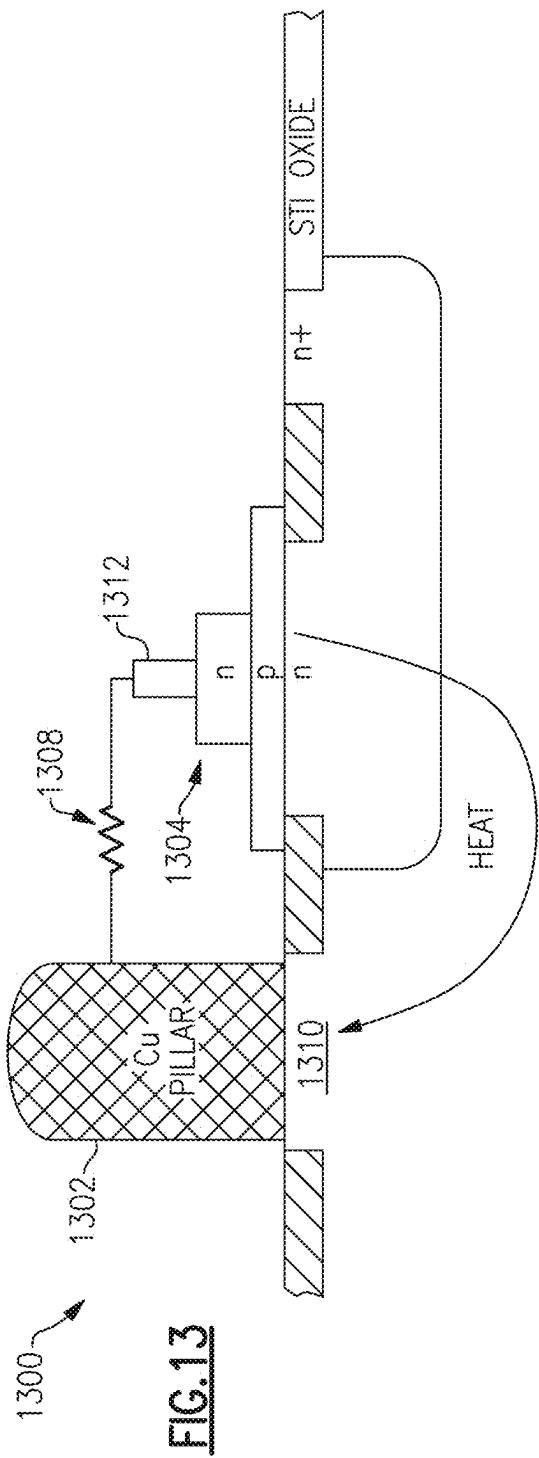


FIG. 12



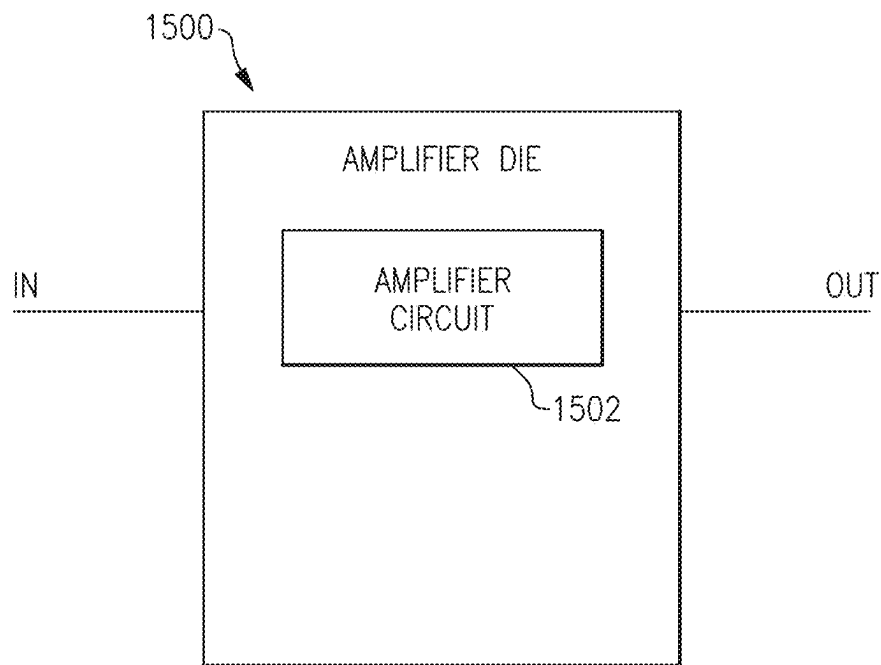


FIG.15

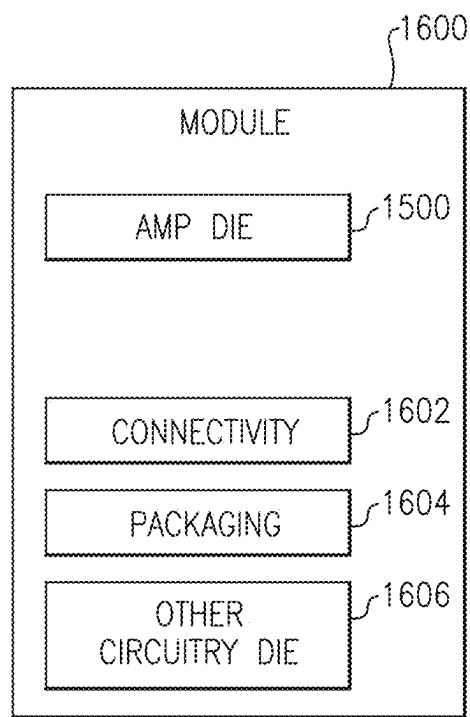


FIG.16

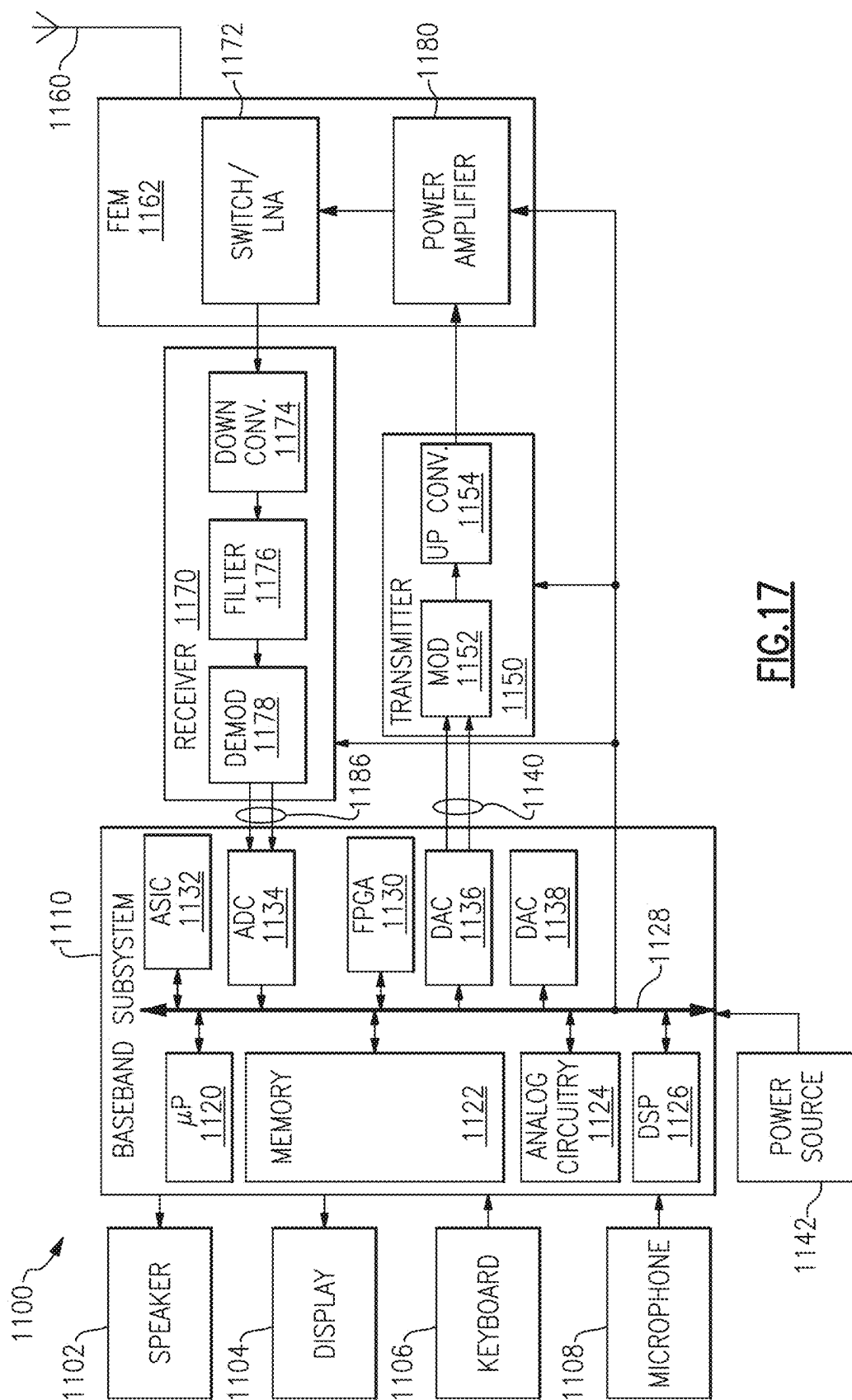


FIG.17

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DIRECT SUBSTRATE TO SOLDER BUMP CONNECTION FOR THERMAL MANAGEMENT IN FLIP CHIP AMPLIFIERS

INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS

Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 CFR 1.57.

BACKGROUND

Embodiments relate to power-amplifier design, and in particular to managing electrothermal performance in flip-chip power-amplifier design.

SUMMARY

In an embodiment, an alternative bumping method for linear flip-chip SiGe-power-amplifier IC design implements emitter ballasting and achieves equivalent or lower thermal resistance to that obtained with direct emitter bumping and base ballasting.

In another embodiment, the metal bump or pillars are placed adjacent to the NPN transistor arrays that are used in the power amplifier for RF power generation. By placing the metal in intimate contact with the silicon substrate, the heat generated by the NPN arrays flows down into the silicon substrate and then out the metal bump/pillar. Advantageously, the metal bump/pillar also forms an electrical ground connection in close proximity to the NPN array and, consequently, the emitter ballast resistors can be connected to the grounding point.

In yet another embodiment, a method to provide equivalent or lower thermal resistance for a SiGe NPN array of a power amplifier relative to the emitter bumping method is disclosed. In an embodiment, the Cu pillar connects directly to the bulk silicon substrate adjacent to the NPN transistor array. The metal area and contact-via area is maximized to the extent of the bump pad to provide less thermal resistance without increasing the die area. The heat generated in each emitter stripe of the transistor array is more efficiently transferred to the bulk silicon rather than to the polysilicon-emitter-contact via because of the physical differences in the top side versus the backside contact surface area of the integrated circuit (IC) or chip. Since the bulk silicon has relatively high thermal conductivity, the heat generated in the NPN transistor array spreads quickly and is efficiently transferred to a large silicon contact pad and Cu pillar.

In a further embodiment, the Cu pillar bump electrically and thermally connects to the bulk silicon and not to the emitter contact. A ballast resistor can be electrically connected and physically placed between each emitter of the NPN transistor array and the bump pad, which in an embodiment, optimizes the electrothermal configuration or layout for a linear SiGe power amplifier.

According to a number of embodiments, the disclosure relates to a method to implement an emitter-ballasted amplifier in a flip chip configuration. The method comprises forming at least one transistor over a silicon substrate, forming a metal pillar with respect to the silicon substrate such that a bottom of the metal pillar is in direct contact with the silicon substrate, and forming a first resistor in electrical communication with an emitter of the at least one transistor and with the metal pillar, heat generated during operation of

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the emitter-ballasted amplifier being transferred through the silicon substrate to the bottom of the metal pillar. In an embodiment, the method further comprises forming the metal pillar in a cavity etched into the silicon substrate such that the metal pillar protrudes outwards and upwards from the cavity, the bottom and at least a portion of the side of the metal pillar in direct contact with the silicon substrate.

In an embodiment, the metal pillar is configured to provide a flip chip interconnection for the emitter-ballasted amplifier. In another embodiment, the metal pillar is configured to form a part of a thermal path for heat generated by the at least one transistor when the emitter-ballasted amplifier is operating. In a further embodiment, the metal pillar does not extend over the base of the at least one transistor. In a yet further embodiment, the metal pillar does not extend over any significant portion of the at least one transistor.

In an embodiment, the metal pillar is configured to provide a ground connection. In another embodiment, the method further comprises forming a second resistor in communication with a base of the at least one transistor. In a further embodiment, the metal pillar includes copper. In a yet further embodiment, the metal pillar forms at least one solder bump of a flip chip.

In an embodiment, the at least one transistor includes an NPN bipolar junction transistor. In another embodiment, the emitter-ballasted amplifier includes a SiGe power amplifier.

Certain embodiments relate to an emitter-ballasted amplifier comprising at least one transistor formed over a silicon substrate, a metal pillar formed with respect to the silicon substrate such that a bottom of the metal pillar is in direct contact with the silicon substrate, and a first resistor in communication with an emitter of the at least one transistor and with the metal pillar, heat generated during operation of the at least one transistor being transferred through the silicon substrate to the bottom of the metal pillar. In an embodiment, the emitter-ballasted amplifier further comprises a cavity etched into the silicon substrate, the metal pillar formed within the cavity such that the metal pillar protrudes outwards and upwards from the cavity, the bottom and at least a portion of the side of the metal pillar in direct contact with the silicon substrate.

In an embodiment, the metal pillar is configured to provide a flip chip interconnection. In another embodiment, the metal pillar is configured to form a part of a thermal path for heat generated by the at least one transistor. In a further embodiment, the metal pillar is configured to provide a ground connection. In a yet further embodiment, the metal pillar forms at least one solder bump of a flip chip.

According to further embodiments, the disclosure related to a wireless mobile device comprising an antenna configured to receive and transmit radio frequency (RF) signals, a transmit/receive switch configured to pass an amplified RF signal to the antenna for transmission, and a multi-chip module including a flip chip amplifier die that includes at least one emitter-ballasted amplifier configured to amplify an RF input signal and to generate the amplified RF signal, where the at least one emitter-ballasted amplifier includes at least one transistor formed over a silicon substrate, a metal pillar formed with respect to the silicon substrate such that a bottom of the metal pillar is in direct contact with the silicon substrate, and a first resistor in communication with an emitter of the at least one transistor and the metal pillar, and an output matching network die including an output matching network circuit configured to match an impedance of a fundamental frequency of the amplified RF signal. In an embodiment, the at least one emitter-ballasted amplifier

further includes a second resistor in communication with a base of the at least one transistor to provide base-ballasting.

According to a number of embodiments, the disclosure relates to an amplifier die comprising an emitter-ballasted amplifier including at least one transistor formed over a silicon substrate, a metal pillar formed with respect to the silicon substrate such that a bottom of the metal pillar is in direct contact with the silicon substrate, and a first resistor in communication with an emitter of the at least one transistor and with the metal pillar, where heat generated during operation of the at least one transistor is transferred through the silicon substrate to the bottom of the metal pillar. The amplifier die further comprises an input pad electrically connected to a first terminal of the emitter-ballasted amplifier, an output pad electrically connected to a second terminal of the emitter-ballasted amplifier, and a plurality of interconnections configured to electrically connect at least the first resistor to the metal pillar. In an embodiment, a portable transceiver comprises the amplifier die.

Certain embodiments relate to a multi-chip module comprising a flip chip amplifier die including at least one emitter-ballasted amplifier configured to amplify an input signal and to generate an amplified output signal, where the at least one emitter-ballasted amplifier includes at least one transistor formed over a silicon substrate, a metal pillar formed with respect to the silicon substrate such that a bottom of the metal pillar is in direct contact with the silicon substrate, and a first resistor in communication with an emitter of the at least one transistor and with the metal pillar. Heat generated during operation of the at least one transistor is transferred through the silicon substrate to the bottom of the metal pillar. The multi-chip module further comprises an output matching network die including an output matching network circuit configured to match an impedance of a fundamental frequency of the amplified output signal. In an embodiment, the emitter-ballasted amplifier includes a SiGe power amplifier.

Advantages of embodiments disclosed herein include:

1. An efficient method to remove heat from a SiGe NPN power transistor array which results in equivalent or lower thermal resistance than emitter bumping, thereby improving power, gain, and linearity of the SiGe power amplifier.
2. Emitter ballasting instead of base ballasting for thermal stability and improved linearity of a SiGe power amplifier.
3. An efficient design of collector and base power combining networks. When emitter bumps are used, the collector and base of the transistor exit the array orthogonally to the emitter, which creates unequal phase distribution between the base and collector or equal phase with less optimum impedance transformation networks from additional routing.
4. A straightforward conversion from chip-on-board (COB) and/or through-silicon-via (TSV) to flip chip (FC) package design and vice versa due to fewer changes to matching network layout and the overall IC floor plan.
5. A low impedance substrate connection, which achieves optimum linearity, output power, and efficiency from the NPN transistor array.
6. Superior linearity and ruggedness due to the use of emitter ballasting, as base ballasting degrades amplifier performance due to the reduction in collector to emitter breakdown voltage (ruggedness), reduction in peak current (V_{be} droop), and increased base band impedance which degrades linearity (memory effect).

7. Prevents thermal coupling between adjacent arrays. The emitter bump removes heat that would otherwise be transferred to adjacent transistor arrays. Removal of the heat reduces current mismatch conditions, limits the amount of current collapse, and reduces thermal run away of the transistor arrays, which increases the safe operating area, increases reliability, and increases linearity of the amplifier.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates an exemplary amplifying circuit, according to certain embodiments.

FIG. 2 is a schematic illustrating a base-ballasted amplifying circuit, according to certain embodiments.

FIG. 3 is a schematic illustrating an emitter-ballasted amplifying circuit, according to certain embodiments.

FIG. 4 illustrates an exemplary flip-chip interconnection of semiconductor devices, according to certain embodiments.

FIG. 5 illustrates a top view of an exemplary flip-chip amplifier layout for a base-ballasted amplifier, according to certain embodiments.

FIG. 6 illustrates a cross section of the base-ballasted amplifier of FIG. 5, according to certain embodiments.

FIG. 7 illustrates a top view of an exemplary flip-chip amplifier layout for an emitter-ballasted amplifier, according to certain embodiments.

FIG. 8 illustrates a cross section of the emitter-ballasted amplifier of FIG. 7, according to certain embodiments.

FIG. 9 is a schematic illustrating a base and emitter-ballasted amplifying circuit, according to certain embodiments.

FIG. 10 illustrates a top view of an exemplary flip-chip amplifier layout for a base and emitter-ballasted amplifier, according to certain embodiments.

FIG. 11 illustrates a cross section of the base and emitter-ballasted amplifier of FIG. 10, according to certain embodiments.

FIG. 12 illustrates a perspective view of an exemplary amplifier layout showing Cu pillars beside transistors arrays, according to certain embodiments.

FIG. 13 illustrates another embodiment of electrothermal design for emitter-ballasted amplifiers implemented in flip chip configuration, according to certain embodiments.

FIG. 14 illustrates another embodiment of electrothermal design for emitter-ballasted amplifiers implemented in flip chip configuration, according to certain embodiments.

FIG. 15 is an exemplary block diagram of an amplifier die, according to certain embodiments.

FIG. 16 is an exemplary block diagram of an amplifier module, according to certain embodiments.

FIG. 17 is an exemplary block diagram illustrating a simplified portable transceiver including embodiments of flip-chip power amplifier layouts, according to certain embodiments.

DETAILED DESCRIPTION

FIG. 1 is a schematic of an exemplary amplifying circuit 100 comprising an amplifier 102 that is configured to amplify an input signal V_{in} to provide an amplified output signal V_{out} . In an embodiment, the amplifier 102 comprises a power amplifier. In an embodiment, an output matching network receives the amplified output signal V_{out} and matches an impedance of a fundamental frequency of the

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amplified output signal. In an embodiment, a front end module comprises the amplifier **102** and the output matching network.

In the design of bipolar power amplifier devices, ballasting is employed to limit the amount of current in the device to maintain thermal stability and achieve good electrical performance. In an embodiment, ballasting resistors electrically couple to the individual NPN cells that make up a power transistor array. Either base or emitter ballasting can prevent thermal runaway and catastrophic failure of the NPN array.

FIG. **2** illustrates an embodiment of a base-ballasted amplifier **200** comprising an array of 1 to N base-ballasted transistor pairs **202(1)-202(n)**. In an embodiment, each base-ballasted transistor pair **202(n)** comprises a first transistor **Q1n**, a second transistor **Q2n**, where each transistor **Q1n**, **Q2n** has a base, an emitter, and a collector. Each base-ballasted transistor pair **202(n)** further comprises a first base resistor **Rb1n**, a second base resistor **Rb2n**, a first base capacitor **Cb1n**, and a second base capacitor **Cb2n**. In an embodiment, the amplifier **200** comprises a power amplifier. In a further embodiment, the amplifier **200** comprises a SiGe power amplifier.

For each base-ballasted transistor pair **202(n)**, a base terminal of the first transistor **Q1n** electrically couples to a first terminal of the first base resistor **Rb1n** and a first terminal of the first base capacitor **Cb1n**. And a base terminal of the second transistor **Q2n** electrically couples to a first terminal of the second base resistor **Rb2n** and a first terminal of the second base capacitor **Cb2n**.

Second terminals of the first base capacitors **Cb11** through **Cb1n** electrically couple to second terminals of the second base capacitors **Cb21** through **Cb2n** and to each other to form an amplifier base or an input **204** to the base-ballasted amplifier **200**. In an embodiment, the input **204** comprises an RF input.

Second terminals of the first base resistors **Rb11** through **Rb1n** electrically couple to second terminals of the second base resistors **Rb21** through **Rb2n** and to each other and are in communication with a DC base signal **206**. In an embodiment, the DC base signal **206** comprises a ground signal.

Collector terminals the first transistors **Q11** through **Q1n** electrically couple to collector terminals of the second transistors **Q21** through **Q2n** and to each other to form an amplifier collector or an output **208** from the base-ballasting amplifier **200**. In an embodiment, the output **208** comprises an RF output.

Emitter terminals the first transistors **Q11** through **Q1n** electrically couple to emitter terminals of the second transistors **Q21** through **Q2n** and to each other to form an amplifier emitter **210** of the base-ballasting amplifier **200**.

FIG. **3** illustrates an embodiment of an emitter-ballasted amplifier **300** comprising an array of 1 to N emitter-ballasted transistor pairs **302(1)-302(n)**. In an embodiment, each emitter-ballasted transistor pair **302(n)** comprises the first transistor **Q1n**, the second transistor **Q2n**, where each transistor **Q1n**, **Q2n** has a base, an emitter, and a collector. Each emitter-ballasted transistor pair **302(n)** further comprises a first emitter resistor **Re1n**, a second emitter resistor **Re2n**, the first base capacitor **Cb1n**, and the second base capacitor **Cb2n**. In an embodiment, the amplifier **300** comprises a power amplifier. In a further embodiment, the amplifier **300** comprises a SiGe power amplifier.

For each emitter-ballasted transistor pair **302(n)**, the emitter terminal of the first transistor **Q1n** electrically couples to a first terminal of the first emitter resistor **Re1n**. And the emitter terminal of the second transistor **Q2n** electrically

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couples to a first terminal of the second emitter resistor **Re2n**. Second terminals of the first emitter resistors **Re11** through **Re1n** electrically couple to second terminals of the second emitter resistors **Re21** through **Re2n** and to each other to form an amplifier emitter **310** of the emitter-ballasted amplifier **300**.

Base terminals of the first transistors **Q11** through **Q1n** electrically couple to base terminals of the second transistors **Q21** through **Q2n**, to first terminals of the first base capacitors **Cb11** through **Cb1n**, and to first terminals of the second base capacitors **Cb21** through **Cb2n** and are in communication with a DC base signal **306**. In an embodiment, the DC base signal **306** comprises a ground signal.

Second terminals of the first base capacitors **Cb11** through **Cb1n** electrically couple to second terminals of the second base capacitors **Cb21** through **Cb2n** and to each other to form an amplifier base or an input **304** to the emitter-ballasted amplifier **300**. In an embodiment, the input **304** comprises an RF input.

Collector terminals the first transistors **Q11** through **Q1n** electrically couple to collector terminals of the second transistors **Q21** through **Q2n** and to each other to form an amplifier collector or an output **308** from the emitter-ballasting amplifier **300**. In an embodiment, the output **308** comprises an RF output.

For linear SiGe power amplifiers, emitter ballasting is typically preferred over base ballasting because base ballasting has many adverse effects on performance, such as but not limited to, lower breakdown voltage and poorer linearity. Emitter ballasting not only provides thermal stability but also achieves better electrical performance than base ballasting.

Flip chip, also known as controlled collapse chip connection or its acronym, C4, is a method for interconnecting semiconductor devices, such as integrated circuits or IC chips and microelectromechanical systems (MEMS), to external circuitry with solder bumps that have been deposited onto bump pads.

FIG. **4** illustrates an exemplary flip-chip assembly **400** comprising an integrated circuit (IC) **410**. In an embodiment, the IC **410** comprises an amplifier, such as, but not limited to amplifier **200**, **300**. In another embodiment, the IC **410** comprises a power amplifier. In a further embodiment, the IC **410** comprises a SiGe power amplifier.

Copper (Cu) pillars or solder bumps **402** are deposited on Cu solder pads or bump pads **404** on the top side of the wafer associated with the IC **410** during the final wafer-processing step. In order to mount the IC **410** to external circuitry **406** (e.g., a circuit board or another chip or wafer), it is flipped over so that its top side faces down, and aligned so that its Cu solder pads **404** align with matching pads **408** on the external circuitry **406**, and then the solder is reflowed to complete the interconnect. This is in contrast to wire bonding, in which an integrated circuit is mounted upright, and wires are used to interconnect chip pads to the external circuitry **406**.

In power amplifier design, it is important to minimize the thermal resistance and limit the negative effects of excess heat on circuit performance parameters, such as but not limited to power, gain, and linearity. In flip-chip power-amplifier design, the amplifier emitters **210**, **310** often connect directly to a copper (Cu) pillar pad associated with a Cu pillar, which provides heat sinking for heat dissipation.

FIG. **5** illustrates a top view of an exemplary flip-chip amplifier layout for electrothermal management of a base-ballasted amplifier **500**. A Cu pillar **502** sits over the

amplifier **500**, which comprises an array **504** of transistors **Q11-Q1_n**, **Q21-Q2_n** and a plurality of base-ballasting resistors **506**.

FIG. **6** illustrates a cross section **600** of the base-ballasted amplifier **500** of FIG. **5**. FIG. **6** shows base-ballasted transistors **Q1**, **Q2** of the array **504** formed over a bulk silicon substrate **610**. The base terminal of the transistor **Q1**, **Q2** is in communication with the base-ballasting resistor **506**. FIG. **6** further shows the emitter of the transistor **Q1**, **Q2** in communication with an emitter-contact via **602**. In an embodiment, the emitter-contact via **602** comprises a polysilicon-emitter-contact via. Inter-level metals and contacts **608**, in an embodiment, comprise L3 metal, inter-level contacts (ILC), and L2 metal, and are in communication with the emitter contact **602** and the Cu pillar **502**. The Cu pillar **502**, through a Cu solder pad, is in contact with a laminate interposer or printed circuit board **606**. The laminate interposer or printed circuit board **606** comprises the Cu solder pad, a Cu via, and a Cu heat sink/ground plane **604**.

Referring to FIGS. **5** and **6**, in an embodiment, a solution for thermal management in flip-chip power-amplifier design comprises connecting the emitter-contact via **602** to the Cu pillar **502** using all metal levels and contact-via levels **608** available in the technology. The Cu pillar **502** then connects to the heat sink/ground **604** on the laminate interposer or printed circuit board **606**. Heat is removed from the emitter through relatively small area polysilicon-emitter contact vias **602**, which transition up through the inter-level metals and contacts **608** to the Cu pillar **502** and to the heat sink/ground plane **604**. The emitter contact area is substantially less than that achieved with silicon bumping, which results in restricted heat flow.

For example, the Cu pillar **502**, which can measure approximately $90\text{ }\mu\text{m}\times 180\text{ }\mu\text{m}$, is transitioned and reduced down to the size of the polysilicon-emitter-contact via **602** which, in an embodiment, is approximately $25\text{ }\mu\text{m}\times 0.4\text{ }\mu\text{m}$ or less. In certain embodiments, there may be approximately 60 or more emitter contacts **602** made to a single Cu pillar **502**. The area of the inter-level metals and contacts **608** is also constricted due to the presence of the collector terminals and the desire to make robust electrical contact to the collector terminals. This reduction in metal and contact area increases thermal resistance.

While emitter ballasting would improve electrical properties of the amplifier **500**, applying emitter ballasting may be impractical to implement and it may lead to an excessive rise in the thermal resistance and junction temperature due to the discontinuity caused by the emitter-ballasting resistor in the heat transfer path. Thus, base ballasting is used to provide thermal stability since emitter ballasting is neither possible nor desirable without causing significant degradation in the thermal conductivity.

For at least these reasons, connecting the polysilicon emitter directly to the Cu pillar **502** and heat sink **604** with base ballasting, instead of the more electrically optimal emitter ballasting, has traditionally been the electrothermal solution for flip-chip power-amplifier design.

Embodiments of better electrothermal design for amplifiers implemented in flip chip configuration are disclosed herein. In an embodiment, the amplifiers comprise power amplifiers. In another embodiment, the amplifiers comprise SiGe power amplifiers.

FIG. **7** illustrates a top view of an exemplary flip-chip amplifier layout for electrothermal management of an emitter-ballasted amplifier **700** that can be used for RF power

generation. Amplifier **700** comprises an array **704** of transistors **Q11-Q1_n**, **Q21-Q2_n** and a plurality of emitter-ballasting resistors **708**.

FIG. **8** illustrates a cross section of the emitter-ballasted amplifier **700** of FIG. **7**. FIG. **8** shows emitter-ballasted transistors **Q1**, **Q2** of the array **704** formed over a bulk silicon substrate **810**. The emitter terminal of the transistor **Q1**, **Q2** is in communication with the emitter-ballasting resistor **706**, which introduces a discontinuity in the heat transfer of the heat generated in the emitter to the Cu pillar **702**. FIG. **8** further shows inter-level metals and contacts **808** in communication with the bulk silicon substrate **810** and the Cu pillar **702**. In an embodiment, the inter-level metals and contacts **808** comprise L3 metal, inter-level contacts (ILC), L2 metal, and L1 metal. The Cu pillar **702**, through a Cu solder pad, is in contact with the laminate interposer or printed circuit board **806**. The laminate interposer or printed circuit board **806** comprises the Cu solder pad, a Cu via, and a Cu heat sink/ground plane **804**.

Referring to FIGS. **7** and **8**, in an embodiment, a solution for thermal management in flip-chip power-amplifier design comprises placing the metal bumps or Cu pillars **702** adjacent to the transistor arrays **704** and over inter-level metals and contacts **808**. By placing the metal of the Cu pillar **702** and the inter-level metals and contacts **808** in intimate contact with the bulk silicon substrate **810**, the heat generated by the transistor arrays **704** flows into the silicon substrate **810**, through the inter-level metals and contacts **808**, and then out the Cu pillar **702** which is in thermal contact with the heat sink/ground plane **804** of the laminate interposer or printed circuit board **806**.

Thus, the thermal properties of the silicon substrate **810** spread the heat to the substrate contacts, such as the inter-level metals and contacts **808**. The heat is then removed through the large area of the substrate contacts, to the Cu pillars **702** and then to the heat sink/ground plane **804**.

The substrate contacts provided by the inter-level metals and contacts **808** and the Cu pillars **702** also limit mutual heating of adjacent transistor arrays **704**, which improves many electrical characteristics, such as current uniformity, gain versus time characteristics, and the like, of the power amplifier **700**.

In another embodiment, the metal bump or Cu pillar **702** forms an electrical ground connection to the ground plane **804**, which in turn forms a grounding point through the inter-level metals and contacts **808**. The Cu pillar **702** is in close proximity to the transistor array **704** and, the emitter-ballasting resistors **708** can be in communication with the grounding point. In an embodiment, the grounding point provides emitter degeneration inductance to maintain high gain over a temperature range. In an embodiment, the ground is located on the interposed or printed circuit board (PCB) **406**.

FIG. **9** illustrates an embodiment of a base and emitter-ballasted amplifier **900** comprising an array of 1 to N base and emitter-ballasted transistor pairs **902(1)-902(n)**. In an embodiment, each base and emitter-ballasted transistor pair **902(n)** comprises a first transistor **Q1_n**, a second transistor **Q2_n**, where each transistor **Q1_n**, **Q2_n** has a base, an emitter, and a collector. Each base and emitter-ballasted transistor pair **902(n)** further comprises the first base resistor **Rb1_n**, the second base resistor **Rb2_n**, the first base capacitor **Cb1_n**, the second case capacitor **Cb2_n**, the first emitter resistor **Re1_n**, and the second emitter resistor **Re2_n**. In an embodiment, the amplifier **900** comprises a power amplifier. In a further embodiment, the amplifier **900** comprises a SiGe power amplifier.

For each base and emitter-ballasted transistor pair **902**(*n*), the base terminal of the first transistor **Q1_n** electrically couples to the first terminal of the first base resistor **Rb1_n** and the first terminal of the first base capacitor **Cb1_n**. And the base terminal of the second transistor **Q2_n** electrically couples to the first terminal of the second base resistor **Rb2_n** and the first terminal of the second base capacitor **Cb2_n**.

Second terminals of the first base capacitors **Cb11** through **Cb1_n** electrically couple to second terminals of the second base capacitors **Cb21** through **Cb2_n** and to each other to form an amplifier base or an input **904** to the base and emitter-ballasted amplifier **900**. In an embodiment, the input **904** comprises an RF input.

Second terminals of the first base resistors **Rb11** through **Rb1_n** electrically couple to second terminals of the second base resistors **Rb21** through **Rb2_n** and to each other and are in communication with a DC base signal **906**. In an embodiment, the DC base signal **906** comprises a ground signal.

For each base and emitter-ballasted transistor pair **902**(*n*), the emitter terminal of the first transistor **Q1_n** electrically couples to the first terminal of the first emitter resistor **Re1_n**. And the emitter terminal of the second transistor **Q2_n** electrically couples to the first terminal of the second emitter resistor **Re2_n**. Second terminals of the first emitter resistors **Re11** through **Re1_n** electrically couple to second terminals of the second emitter resistors **Re21** through **Re2_n** and to each other to form an amplifier emitter **910** of the base and emitter-ballasted amplifier **900**.

Collector terminals of the first transistors **Q11** through **Q1_n** electrically couple to collector terminals of the second transistors **Q21** through **Q2_n** and to each other to form an amplifier collector or an output **908** from the base and emitter-ballasting amplifier **900**. In an embodiment, the output **908** comprises an RF output.

In an embodiment, the base-ballasting resistor provides base resistance for the base and emitter-ballasted amplifier **900**. The base resistance provides RF isolation and assists with low frequency stability for the amplifier **900** under mismatched termination conditions.

FIG. 10 illustrates a top view of an exemplary flip-chip amplifier layout for electrothermal management of a base and emitter-ballasted amplifier **1000**. Amplifier **1000** comprises an array **1004** of transistors **Q11-Q1_n**, **Q21-Q2_n**, a plurality of base-ballasting resistors **1006**, and a plurality of emitter-ballasting resistors **1008**.

FIG. 11 illustrates a cross section **1700** of the base and emitter-ballasted amplifier **1000** of FIG. 10. FIG. 11 shows base and emitter-ballasted transistors **Q1**, **Q2** of the array **1004** formed over a bulk silicon substrate **1710**. The base terminal of the transistor **Q1**, **Q2** is in communication with the base-ballasting resistor **1006**. The emitter terminal of the transistor **Q1**, **Q2** is in communication with the emitter-ballasting resistor **1008**, which introduces a discontinuity in the heat transfer of the heat generated in the emitter to the Cu pillar **1002**. FIG. 11 further shows inter-level metals and contacts **1708** in communication with the bulk silicon substrate **1710** and the Cu pillar **1002**. In an embodiment, the inter-level metals and contacts **1708** comprise L3 metal, inter-level contacts (ILC), L2 metal, and L1 metal. The Cu pillar **1002**, through a Cu solder pad, is in contact with a laminate interposer or printed circuit board **1706**. The laminate interposer or printed circuit board **1706** comprises the Cu solder pad, a Cu via, and a Cu heat sink/ground plane **1704**.

Referring to FIGS. 10 and 11, in an embodiment, a solution for thermal management in flip-chip power-amplifier design comprises placing metal bumps or Cu pillars

1002 adjacent to the transistor arrays **1004** and over inter-level metals and contacts **1708**. By placing the metal of the Cu pillar **1002** and the inter-level metals and contacts **1708** in intimate contact with a silicon substrate **1710**, the heat generated by the transistor arrays **1004** flows into the silicon substrate **1710**, through the inter-level metals and contacts **1708**, and then out the Cu pillar **1002** which is in thermal contact with the heat sink/ground plane **1704** of the laminate interposer or printed circuit board **1706**.

Thus, as described above, the thermal properties of the silicon substrate **1710** spread the heat to the substrate contacts, such as the inter-level metals and contacts **1708**. The heat is then removed through the large area of the substrate contacts, to the Cu pillars **1002** and the heat sink/ground plane **1704**.

In another embodiment, the metal bump or Cu pillar **1002** forms an electrical ground connection to the ground plane **1704**, which in turn forms a grounding point through the inter-level metals and contacts **1708**. The Cu pillar **1002** is in close proximity to the transistor array **1004** and, the emitter-ballasting resistors **1008** can be in communication with the grounding point.

FIG. 12 illustrates a perspective view of an exemplary flip chip layout of an amplifier **1200** showing at least Cu pillars **1202** beside transistor arrays **1204**. In an embodiment, the amplifier **1200** comprises at least one of a power amplifier, a SiGe power amplifier, an emitter-ballasted amplifier, a base and emitter-ballasted amplifier, and a base-ballasted amplifier. FIG. 12 further illustrates the current uniformity benefits of the design. The three copper pillars **1202** are arranged with two transistor arrays **1204** between the left copper pillar **1202** and the middle copper pillar **1202** and with another two transistor arrays **1204** between the middle copper pillar **1202** and the right copper pillar **1202**. The middle pillar **1202** provides thermal sharing of the heat generated by the transistor arrays **1204** on both sides of it. The thermal sharing promotes uniform current density between all of the transistor arrays **1204**. When the currents are uniform, the amplifier **1200** exhibits more linear performance than when the currents are not uniform.

FIGS. 13 and 14 illustrates additional embodiments of electrothermal design for emitter-ballasted amplifiers implemented in flip chip configuration.

FIG. 13 illustrates an amplifier layout of an amplifier **1300** comprising NPN arrays **1304**, a bump or Cu pillar **1302**, and an emitter-ballasting resistor **1308** in communication with an emitter **1312** and the Cu pillar **1302**. The Cu pillar **1302** sets directly on an active silicon region **1310** and adjacent to the NPN arrays **1304** to provide a more efficient thermal path than when the back-end-of-line (BEOL) stack of metals and dielectrics are underneath the Cu pillar **1302**. The BEOL stack is not as thermally conductive as the solid metal forming the bump or Cu pillar **1302**. By removing the BEOL stack from underneath the Cu pillar **1302**, the layout embodiment illustrated in FIG. 13 provides less thermal resistance to the thermal path than when the BEOL stack is positioned underneath the Cu pillar **1302**.

FIG. 14 illustrates an amplifier layout of an amplifier **1400** comprising NPN arrays **1404**, a bump or Cu pillar **1402**, and an emitter-ballasting resistor **1408** in communication with an emitter **1412** and the Cu pillar **1402**. In this layout embodiment, the Cu pillar **1402** is sunk into a silicon active region **1410** and is adjacent to the NPN arrays **1404**. In an embodiment, a cavity is etched into the plane of the wafer. The metal bump or Cu pillar **1402** is formed within the

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cavity and protrudes outwards and upwards to provide additional thermal paths along the sides of the Cu pillar 1402 for waste heat.

FIG. 15 illustrates an exemplary block diagram of an amplifier die 1500 including an embodiment of an amplifier 1502. In an embodiment, the amplifier 1502 comprises the amplifier circuit or amplifier layout 102, 200, 300, 500, 600, 700, 800, 900, 1000, 1200, 1300, 1400, 1700.

FIG. 16 illustrates an exemplary block diagram of a module 1600 including amplifier die 1500 of FIG. 15. The module 1600 further includes connectivity 1602 to provide signal interconnections, packaging 1604, such as for example, a package substrate, for packaging of the circuitry, and other circuitry die 1606, such as, for example amplifiers, pre-filters, post filters modulators, demodulators, down converters, and the like, as would be known to one of skill in the art of semiconductor fabrication in view of the disclosure herein. In an embodiment, the module 1600 comprises a front-end module.

FIG. 17 illustrates an exemplary block diagram illustrating a simplified portable transceiver 1100 including an embodiment of the amplifier 102, 200, 300, 500, 600, 700, 800, 900, 1000, 1200, 1300, 1400, 1500, 1600, 1700.

The portable transceiver 1100 includes a speaker 1102, a display 1104, a keyboard 1106, and a microphone 1108, all connected to a baseband subsystem 1110. A power source 1142, which may be a direct current (DC) battery or other power source, is also connected to the baseband subsystem 1110 to provide power to the portable transceiver 1100. In a particular embodiment, portable transceiver 1100 can be, for example but not limited to, a portable telecommunication device such as a mobile cellular-type telephone. The speaker 1102 and the display 1104 receive signals from baseband subsystem 1110, as known to those skilled in the art. Similarly, the keyboard 1106 and the microphone 1108 supply signals to the baseband subsystem 1110.

The baseband subsystem 1110 includes a microprocessor (μ P) 1120, memory 1122, analog circuitry 1124, and a digital signal processor (DSP) 1126 in communication via bus 1128. Bus 1128, although shown as a single bus, may be implemented using multiple busses connected as necessary among the subsystems within the baseband subsystem 1110. The baseband subsystem 1110 may also include one or more of an application specific integrated circuit (ASIC) 1132 and a field programmable gate array (FPGA) 1130.

The microprocessor 1120 and memory 1122 provide the signal timing, processing, and storage functions for portable transceiver 1100. The analog circuitry 1124 provides the analog processing functions for the signals within baseband subsystem 1110. The baseband subsystem 1110 provides control signals to a transmitter 1150, a receiver 1170, and a power amplifier 1180, for example.

It should be noted that, for simplicity, only the basic components of the portable transceiver 1100 are illustrated herein. The control signals provided by the baseband subsystem 1110 control the various components within the portable transceiver 1100. Further, the function of the transmitter 1150 and the receiver 1170 may be integrated into a transceiver.

The baseband subsystem 1110 also includes an analog-to-digital converter (ADC) 1134 and digital-to-analog converters (DACs) 1136 and 1138. In this example, the DAC 1136 generates in-phase (I) and quadrature-phase (Q) signals 1140 that are applied to a modulator 1152. The ADC 1134, the DAC 1136, and the DAC 1138 also communicate with the microprocessor 1120, the memory 1122, the analog circuitry 1124, and the DSP 1126 via bus 1128. The DAC

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1136 converts the digital communication information within baseband subsystem 1110 into an analog signal for transmission to the modulator 1152 via connection 1140. Connection 1140, while shown as two directed arrows, includes the information that is to be transmitted by the transmitter 1150 after conversion from the digital domain to the analog domain.

The transmitter 1150 includes the modulator 1152, which modulates the analog information on connection 1140 and provides a modulated signal to upconverter 1154. The upconverter 1154 transforms the modulated signal to an appropriate transmit frequency and provides the upconverted signal to the power amplifier 1180. The power amplifier 1180 amplifies the signal to an appropriate power level for the system in which the portable transceiver 1100 is designed to operate. In an embodiment, the power amplifier 1180 comprises the amplifier module 1600.

Details of the modulator 1152 and the upconverter 1154 have been omitted, as they will be understood by those skilled in the art. For example, the data on connection 1140 is generally formatted by the baseband subsystem 1110 into in-phase (I) and quadrature (Q) components. The I and Q components may take different forms and be formatted differently depending upon the communication standard being employed.

A front-end module 1162 comprises the power amplifier (PA) circuit 1180 and a switch/low noise amplifier (LNA) circuit 1172. In an embodiment, the switch/low noise amplifier circuit 1172 comprises an antenna system interface that may include, for example, a diplexer having a filter pair that allows simultaneous passage of both transmit signals and receive signals, as known to those having ordinary skill in the art.

In an embodiment, the front-end module 1162 comprises the module 1600. In an embodiment, the amplifier circuit comprises the amplifier 102, 200, 300, 500, 600, 700, 800, 900, 1000, 1200, 1300, 1400, 1500, 1600, 1700.

The power amplifier 1180 supplies the amplified transmit signal to the switch/low noise amplifier circuit 1172. The transmit signal is supplied from the front-end module 1162 to the antenna 1160 when the switch is in the transmit mode.

A signal received by antenna 1160 will be directed from the switch/low noise amplifier 1172 of the front-end module 1162 to the receiver 1170 when the switch is in the receive mode. The low noise amplifier circuitry 1172 amplifies the received signal.

If implemented using a direct conversion receiver (DCR), the downconverter 1174 converts the amplified received signal from an RF level to a baseband level (DC), or a near-baseband level (approximately 100 kHz). Alternatively, the amplified received RF signal may be downconverted to an intermediate frequency (IF) signal, depending on the application. The downconverted signal is sent to the filter 1176. The filter 1176 comprises a least one filter stage to filter the received downconverted signal as known in the art.

The filtered signal is sent from the filter 1176 to the demodulator 1178. The demodulator 1178 recovers the transmitted analog information and supplies a signal representing this information via connection 1186 to the ADC 1134. The ADC 1134 converts these analog signals to a digital signal at baseband frequency and transfers the signal via bus 1128 to the DSP 1126 for further processing.

The methods and apparatus described herein provide amplifier designs for electrothermal management. In embodiments of the amplifier 102, 200, 300, 700, 800, 900, 1000, 1180, 1200, 1300, 1400, 1500, 1600, 1700, the transistors Q1, Q2, Q11-Q1n, Q21-Q2n comprise NPN bipolar

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junction transistors (BJTs). The amplifiers **102, 200, 300, 500, 600, 700, 800, 900, 1000, 1180, 1200, 1300, 1400, 1500, 1600, 1700** can also be implemented using different technologies, such as, but not limited to SiGe, MOS, PNP BJT, HBT, pHEMT, GaN, Gas, InGaP Gas HBT, MOSFET, SOI, Bulk CMOS, CMOS, and the like.

Terminology

Some of the embodiments described above have provided examples in connection with mobile phones. However, the principles and advantages of the embodiments can be used for any other systems or apparatus that have needs for power amplifier systems.

Such a system or apparatus can be implemented in various electronic devices. Examples of the electronic devices can include, but are not limited to, consumer electronic products, parts of the consumer electronic products, electronic test equipment, etc. Examples of the electronic devices can also include, but are not limited to, memory chips, memory modules, circuits of optical networks or other communication networks, and disk driver circuits. The consumer electronic products can include, but are not limited to, a mobile phone such as a smart phone, a telephone, a television, a computer monitor, a computer, a hand-held computer, a laptop computer, a tablet computer, a personal digital assistant (PDA), a PC card, a microwave, a refrigerator, an automobile, a stereo system, a cassette recorder or player, a DVD player, a CD player, a VCR, an MP3 player, a radio, a camcorder, a camera, a digital camera, a portable memory chip, a washer, a dryer, a washer/dryer, a copier, a facsimile machine, a scanner, a multi-functional peripheral device, a wrist watch, a clock, etc. Further, the electronic devices can include unfinished products.

Unless the context clearly requires otherwise, throughout the description and the claims, the words “comprise,” “comprising,” and the like are to be construed in an inclusive sense, as opposed to an exclusive or exhaustive sense; that is to say, in the sense of “including, but not limited to.” The word “coupled”, as generally used herein, refers to two or more elements that may be either directly connected, or connected by way of one or more intermediate elements. Likewise, the word “connected”, as generally used herein, refers to two or more elements that may be either directly connected, or connected by way of one or more intermediate elements. Additionally, the words “herein,” “above,” “below,” and words of similar import, when used in this application, shall refer to this application as a whole and not to any particular portions of this application. Where the context permits, words in the above Detailed Description using the singular or plural number may also include the plural or singular number respectively. The word “or” in reference to a list of two or more items, that word covers all of the following interpretations of the word: any of the items in the list, all of the items in the list, and any combination of the items in the list.

Moreover, conditional language used herein, such as, among others, “can,” “could,” “might,” “can,” “e.g.,” “for example,” “such as” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or states. Thus, such conditional language is not generally intended to imply that features, elements and/or states are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without

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author input or prompting, whether these features, elements and/or states are included or are to be performed in any particular embodiment.

The above detailed description of embodiments of the invention is not intended to be exhaustive or to limit the invention to the precise form disclosed above. While specific embodiments of, and examples for, the invention are described above for illustrative purposes, various equivalent modifications are possible within the scope of the invention, as those skilled in the relevant art will recognize. For example, while processes or blocks are presented in a given order, alternative embodiments may perform routines having steps, or employ systems having blocks, in a different order, and some processes or blocks may be deleted, moved, added, subdivided, combined, and/or modified. Each of these processes or blocks may be implemented in a variety of different ways. Also, while processes or blocks are at times shown as being performed in series, these processes or blocks may instead be performed in parallel, or may be performed at different times.

The teachings of the invention provided herein can be applied to other systems, not necessarily the system described above. The elements and acts of the various embodiments described above can be combined to provide further embodiments.

While certain embodiments of the inventions have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosure. Indeed, the novel methods, apparatus, and systems described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions, and changes in the form of the methods and systems described herein may be made without departing from the spirit of the disclosure. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosure.

What is claimed is:

1. A method to implement an emitter-ballasted amplifier in a semiconductor die, the method comprising:
 - forming a first transistor over a semiconductor substrate of a semiconductor die;
 - forming a first metal structure with respect to the semiconductor substrate such that the first metal structure is in thermal communication with the semiconductor substrate, the first metal structure forming part of a first die contact; and
 - forming a first resistor in electrical communication with an emitter of the first transistor and with the first metal structure, heat generated during operation of the first transistor being transferred to the first metal structure.
2. The method of claim 1 further comprising forming the first metal structure in a cavity in the semiconductor substrate such that the first metal structure is partially within the cavity and partially outside of the cavity.
3. The method of claim 1 wherein the first metal structure is configured to form a part of a thermal path for heat generated by the first transistor.
4. The method of claim 1 wherein the first metal structure does not extend over a base of the first transistor.
5. The method of claim 4 wherein the first metal structure does not extend over any portion of the first transistor.
6. The method of claim 1 wherein the first metal structure is configured to provide a ground connection.
7. The method of claim 1 further comprising forming a second resistor in communication with a base of the first transistor.

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8. The method of claim 1 wherein the first metal structure is a metal pillar.

9. The method of claim 1 wherein a footprint defined by a periphery of the first metal structure does not overlap with a footprint defined by a periphery of the first transistor.

10. The method of claim 1 wherein the emitter-ballasted amplifier includes a silicon-germanium power amplifier.

11. The method of claim 1 further comprising:

forming a second transistor over the semiconductor substrate, a base of the second transistor connected to a base of the first transistor;

forming a second metal structure with respect to the semiconductor substrate such that the second metal structure is in thermal communication with the semiconductor substrate, the second metal structure forming part of a second die contact; and

forming a second resistor in electrical communication with an emitter of the second transistor and with the second metal structure, heat generated during operation of the second transistor being transferred through the semiconductor substrate to the second metal structure.

12. The method of claim 1 wherein the first die contact is a flip-chip contact.

13. An semiconductor die comprising:

a semiconductor substrate;

a first transistor formed over the semiconductor substrate;

a first metal structure positioned with respect to the semiconductor substrate such that a first end of the first metal structure is in thermal communication with the semiconductor substrate, the first metal structure forming part of a first die contact; and

a first resistor in electrical communication with an emitter of the first transistor and with the first metal structure, heat generated during operation of the first transistor being transferred the first metal structure.

14. The semiconductor die of claim 13 further comprising a cavity in the semiconductor substrate, the first metal structure formed partially within the cavity and partially outside of the cavity.

15. The semiconductor die of claim 13 wherein the first metal structure is a metal pillar.

16. The semiconductor die of claim 13 wherein a footprint defined by a periphery of the first metal structure does not overlap with a footprint defined by a periphery of the first transistor.

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17. The semiconductor die of claim 13 wherein the first metal structure is configured to provide a ground connection.

18. The semiconductor die of claim 13 further comprising:

a second transistor formed over the semiconductor substrate, a base of the second transistor connected to a base of the first transistor and forming a radio frequency input of an emitter-ballasted amplifier;

a second metal structure positioned with respect to the semiconductor substrate such that the second metal structure is in thermal communication with the semiconductor substrate, the second metal structure forming part of a second contact;

a second resistor in electrical communication with an emitter of the second transistor and with the second metal structure, heat generated during operation of the second transistor being transferred to the second metal structure.

19. The semiconductor die of claim 13 wherein the first metal structure does not extend over a base of the first transistor.

20. The semiconductor die of claim 13 wherein the first metal structure does not extend over any portion of the first transistor.

21. The semiconductor die of claim 13 wherein the first die contact is a flip-chip contact.

22. A wireless mobile device comprising:

an antenna configured to receive and transmit radio frequency signals; and

a multi-chip module including a semiconductor die that includes at least one emitter-ballasted amplifier configured to amplify a radio frequency input signal, the at least one emitter-ballasted amplifier including a semiconductor substrate, a first transistor formed over the semiconductor substrate, a first metal structure positioned with respect to the semiconductor substrate such that the first metal structure is in thermal communication with the semiconductor substrate, a first resistor in electrical communication with an emitter of the first transistor and with the first metal structure, heat generated during operation of the first transistor being transferred to the first metal structure.

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